

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 1

Foundations: Stochastic Processes and ARMA Models

Prof. Pedro Valls Pereira

`pedro.valls@fgv.br`

Topics covered: Stationarity · ACF/PACF · White Noise · MA & AR Processes
ARMA Models · Box–Jenkins Identification · MLE Estimation · Diagnostic Validation ·
Forecasting

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1. Stochastic Processes

A **stochastic process** is a random function indexed by a time set T and a sample space Ω :

$$X : T \times \Omega \longrightarrow \mathbb{R}, \quad (t, \omega) \mapsto X(t, \omega).$$

A specific parametric example is:

$$X(t, \omega) = Y(\omega) \cos[Z(\omega) + t \cdot U(\omega)],$$

where $Y \sim N(0, \sigma_y^2)$, $Z \sim N(0, \sigma_z^2)$, and $U \sim U[-\pi, \pi]$ are mutually independent — a cosine function with a random amplitude, phase, and frequency.

For each time point t we observe only *one realisation* x_t of the random variable X_t . Recovering the full distribution of X_t from a single observation is impossible without further restrictions. Two types of restriction are imposed:

- (i) **Temporal homogeneity** — the statistical structure does not change over time (stationarity);
- (ii) **Limited memory** — distant past observations provide negligible information about the present (ergodicity).

2. Stationarity

2.1. Strict stationarity

Definition 2.1. $\{X_t\}$ is **strictly stationary** if, for every finite collection of indices $\{t_1, \dots, t_n\} \subset T$ and every integer τ ,

$$D(X_{t_1}, \dots, X_{t_n}) = D(X_{t_1+\tau}, \dots, X_{t_n+\tau}).$$

The joint distribution is invariant to time shifts.

2.2. Weak (covariance) stationarity

Definition 2.2. $\{X_t\}$ is **weakly stationary** (or covariance stationary) if:

- (1) $E(X_t) = \mu < \infty, \quad \forall t,$
- (2) $\text{Var}(X_t) = \gamma(0) < \infty, \quad \forall t,$
- (3) $E[(X_{t_1} - \mu)(X_{t_2} - \mu)] = \gamma(t_2 - t_1), \quad \forall t_1, t_2.$

All means and variances are constant, and all covariances depend only on the lag $s = t_2 - t_1$, not on the absolute time. Intuitively, a stationary process *looks the same through any time window* — Valls' window.

3. Autocovariance and Autocorrelation Functions

3.1. Autocovariance function

For a weakly stationary process, the **autocovariance function** (ACF) at lag s is:

$$\gamma(s) = E[(X_t - \mu)(X_{t-s} - \mu)], \quad s = 0, \pm 1, \pm 2, \dots$$

Proposition 3.1. *The autocovariance function satisfies:*

$$\gamma(0) \geq 0, \tag{1}$$

$$|\gamma(s)| \leq \gamma(0) \quad \forall s \in \mathbb{Z} \quad (\text{Cauchy-Schwarz}), \tag{2}$$

$$\gamma(s) = \gamma(-s) \quad (\text{even function}), \tag{3}$$

$$\gamma(\cdot) \text{ is non-negative definite.} \tag{4}$$

3.2. Autocorrelation function (correlogram)

The **autocorrelation function** normalises the autocovariance by the variance:

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)}, \quad s = 0, 1, 2, \dots$$

Plotting $\rho(s)$ against s gives the *correlogram*.

4. White Noise

Definition 4.1. $\{X_t\}$ is a *white noise process* if:

$$E(X_t) = \mu, \quad \text{Var}(X_t) = \sigma^2, \quad \gamma(s) = \begin{cases} \sigma^2 & s = 0, \\ 0 & s \neq 0. \end{cases}$$

Equivalently (with $\mu = 0$): $E(X_t | \mathbf{I}_{t-1}) = 0$, so white noise is completely unpredictable from its own past — the benchmark against which all model residuals are tested.

4.1. Testing for white noise

Individual lags. Under homoskedasticity, $\hat{\rho}_s \stackrel{a}{\sim} N(0, 1/T)$, giving a 95% confidence band of $\pm 1.96/\sqrt{T}$.

Ljung–Box joint test.

$$Q^* = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \stackrel{a}{\sim} \chi_m^2 \quad \text{under } H_0 : \rho(1) = \dots = \rho(m) = 0.$$

Under heteroskedasticity, use the Bartlett or Newey–West formula for $\text{Var}(\hat{\rho}_s)$.

5. Moving Average Processes

5.1. Definition and moments

A moving average model of order q , $MA(q)$, is:

$$X_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \quad u_t \stackrel{\text{iid}}{\sim} (0, \sigma^2).$$

Moments:

$$E(X_t) = \mu, \tag{5}$$

$$\gamma(0) = (1 + \theta_1^2 + \cdots + \theta_q^2)\sigma^2, \tag{6}$$

$$\gamma(s) = \begin{cases} (\theta_s + \theta_{s+1}\theta_1 + \cdots + \theta_q\theta_{q-s})\sigma^2 & 1 \leq s \leq q, \\ 0 & s > q. \end{cases} \tag{7}$$

The ACF of an $MA(q)$ **cuts off** sharply after lag q . The PACF decays geometrically. Spike count in ACF = q .

5.2. Invertibility condition

An $MA(q)$ is **invertible** if all roots of $\theta(z) = 0$ lie outside the unit circle. An invertible $MA(q)$ has an $AR(\infty)$ representation.

5.3. $MA(1)$ worked example

Consider $X_t = u_t + \theta u_{t-1}$:

$$\rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(s) = 0 \text{ for } s \geq 2.$$

For $\theta_1 = -0.5$: $\rho(1) = -0.4$, $\rho(s) = 0$ for $s \geq 2$.
 For $\theta_1 = 0.45$ (EViews simulation): $\rho(1) \approx 0.38$.

5.4. $MA(2)$ worked example

Consider $X_t = u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2}$:

$$\rho(1) = \frac{(\theta_1 + \theta_1\theta_2)}{1 + \theta_1^2 + \theta_2^2}, \tag{8}$$

$$\rho(2) = \frac{\theta_2}{1 + \theta_1^2 + \theta_2^2}, \quad \rho(s) = 0 \text{ for } s \geq 3. \tag{9}$$

For $\theta_1 = -0.5$, $\theta_2 = 0.25$: $\rho(1) \approx -0.476$, $\rho(2) \approx 0.190$.

6. Autoregressive Processes

6.1. Definition

An **autoregressive model of order p** , $AR(p)$, is:

$$X_t = \mu + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + u_t,$$

or compactly using the lag operator $L^i X_t = X_{t-i}$:

$$\phi(L) X_t = \mu + u_t, \quad \phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p.$$

6.2. Stationarity condition

All roots of $\phi(L) = 0$ must lie **outside** the unit circle (equivalently, roots of the characteristic polynomial in z lie *inside* the unit circle).

- $X_t = X_{t-1} + u_t$: characteristic root is 1 (unit root) $\Rightarrow$ non-stationary.
- $X_t = 3X_{t-1} - 2.75X_{t-2} + 0.75X_{t-3} + u_t$: roots are 1, $2/3$, and $2 \Rightarrow$ not all outside unit circle $\Rightarrow$ non-stationary.

6.3. Wold's decomposition theorem

Any stationary process can be expressed as an $MA(\infty)$:

$$X_t = \psi(L) u_t, \quad \psi(L) = \frac{1}{\phi(L)}.$$

This guarantees existence of the variance and autocovariances.

6.4. Moments and Yule–Walker equations

$$E(X_t) = \frac{\mu}{1 - \phi_1 - \cdots - \phi_p}.$$

Autocovariances satisfy the **Yule–Walker equations**:

$$\begin{aligned} \rho(1) &= \phi_1 + \rho(1)\phi_2 + \cdots + \rho(p-1)\phi_p, \\ \rho(2) &= \rho(1)\phi_1 + \phi_2 + \cdots + \rho(p-2)\phi_p, \\ &\vdots \\ \rho(p) &= \rho(p-1)\phi_1 + \cdots + \phi_p. \end{aligned}$$

For a stationary $AR(p)$, the ACF decays *exponentially* (and may oscillate). The PACF cuts off after lag p .

6.5. AR(1) key results

For $X_t = \phi X_{t-1} + u_t$ with $|\phi| < 1$:

$$\text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2}, \quad (10)$$

$$\gamma(s) = \frac{\phi^s \sigma^2}{1 - \phi^2}, \quad (11)$$

$$\rho(s) = \phi^s, \quad s = 0, 1, 2, \dots \quad (12)$$

Population spectrum: $s_X(\omega) = \frac{\sigma^2}{2\pi[1 + \phi^2 - 2\phi \cos(\omega)]}$.

7. The Partial Autocorrelation Function

The **PACF** at lag k , denoted ϕ_{kk} , measures the correlation between X_t and X_{t-k} after removing the linear effects of all intermediate lags $X_{t-1}, \dots, X_{t-k+1}$. It equals the last coefficient in an OLS regression of X_t on $X_{t-1}, \dots, X_{t-k}$.

Model	ACF	PACF
AR(p)	Decays exponentially / oscillates	Cuts off after lag p
MA(q)	Cuts off after lag q	Decays exponentially / oscillates
ARMA(p, q)	Decays after lag q	Decays after lag p

At lag 1: $\phi_{11} = \rho(1)$. At lag 2: $\phi_{22} = (\rho(2) - \rho(1)^2)/(1 - \rho(1)^2)$.

8. ARMA Processes

8.1. Definition

By combining AR(p) and MA(q) components, we obtain:

$$\phi(L) X_t = \mu + \theta(L) u_t,$$

where $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ and $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$.

Properties: $E(u_t) = 0$, $E(u_t^2) = \sigma^2$, $E(u_t u_s) = 0$ for $t \neq s$.

Mean: $E(X_t) = \mu/(1 - \phi_1 - \dots - \phi_p)$.

For lags beyond q the ACF behaves like a pure AR(p); for lags beyond p the PACF behaves like a pure MA(q).

8.2. ARMA(1,1) population spectrum

For $X_t = \phi_1 X_{t-1} + u_t + \theta_1 u_{t-1}$:

$$s_X(\omega) = \frac{\sigma^2}{2\pi} \cdot \frac{1 + \theta_1^2 + 2\theta_1 \cos(\omega)}{1 + \phi_1^2 - 2\phi_1 \cos(\omega)}.$$

8.3. ARIMA models

An $\text{ARIMA}(p, d, q)$ model is an $\text{ARMA}(p, q)$ applied to the d -th difference of X_t . The integrated process has a characteristic root on the unit circle. Researchers first difference the variable d times until stationarity is achieved, then fit an ARMA to the differenced series.

9. The Box–Jenkins Approach: Identification, Estimation, Validation

9.1. Identification

1. Determine the integration order d (unit root tests — Lecture 2).
2. Use ACF/PACF plots to suggest initial p and q .
3. Apply the *parsimony principle*: prefer the simplest model that adequately describes the data.
4. Select the final specification using information criteria over the entire $\text{ARMA}(p, q)$ grid, keeping the estimation sample constant.

If AR and MA parameters are nearly equal — e.g. an $\text{ARMA}(1,1)$ with $\phi \approx \theta$ — the two polynomials share near-common roots and the ACF/PACF looks like white noise. The model collapses to $\text{ARMA}(0,0)$. Always inspect roots after estimation and apply the parsimony principle.

9.2. Estimation

9.2.1. Exact MLE vs. conditional MLE for $\text{AR}(p)$

Using the prediction-error decomposition:

$$D(X_1, \dots, X_T | \theta) = \prod_{t=p+1}^T D(X_t | \mathbf{I}_{t-1}^{t-p}; \theta) \cdot D(X_1, \dots, X_p | \theta).$$

Exact MLE: includes the unconditional density of the first p observations, $D(X_1, \dots, X_p | \theta) \sim N(\mathbf{0}, \sigma^2 V_p)$. Non-linear in parameters; requires numerical iteration.

Conditional MLE (OLS): treats the first p observations as fixed, maximising the conditional log-likelihood. For $\text{AR}(1)$ this gives a closed form:

$$\hat{\phi} = \frac{\sum_{t=2}^T X_t X_{t-1}}{\sum_{t=2}^T X_{t-1}^2}, \quad \hat{\sigma}^2 = \frac{1}{T} \sum_{t=2}^T (X_t - \hat{\phi} X_{t-1})^2.$$

Differences between exact and conditional MLE are negligible for large T .

Exact MLE log-likelihood for AR(1):

$$\log l(\phi; \sigma^2) = -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=2}^T (X_t - \phi X_{t-1})^2 + \frac{1}{2} \ln(1 - \phi^2) - \frac{(1 - \phi^2)}{2\sigma^2} X_1^2.$$

9.2.2. Newton–Raphson iterative procedure

The score equations $q(\hat{\theta}) = 0$ are generally non-linear. The Newton–Raphson update is:

$$\theta_{i+1} = \theta_i - H^{-1}(\theta_i) q(\theta_i),$$

where $q(\theta) = \partial \log l / \partial \theta$ and $H(\theta) = \partial^2 \log l / \partial \theta \partial \theta'$. Stopping criteria: convergence in the log-likelihood, score, or parameter values.

9.2.3. Estimation for MA(q) — backcasting

MA parameters have no closed-form estimator. The *backcasting algorithm*:

- (i) Initialise $\hat{u}_0 = 0$ and a starting value θ_0 .
- (ii) Construct $\hat{u}_t(\theta) = X_t - \theta \hat{u}_{t-1}(\theta)$ recursively.
- (iii) Minimise $\sum_{t=1}^T \hat{u}_t^2(\theta)$ with respect to θ .

9.3. Validation

Test two hypotheses after estimation:

- (i) Residuals are Gaussian white noise.
- (ii) Parameters are constant over the sample.

OLS residuals are heteroskedastic and serially correlated by construction and cannot be used directly. Use **recursive (standardised) residuals** instead.

9.3.1. Recursive least squares

The recursive OLS estimator updates as:

$$\hat{\beta}_{t+1} = \hat{\beta}_t + \lambda_{t+1} v_{t+1} (1 + x'_{t+1} \lambda_{t+1})^{-1},$$

where $\lambda_{t+1} = (X_t' X_t)^{-1} x_{t+1}$ and the *one-step-ahead forecast error* (recursive residual) is:

$$v_{t+1} = y_{t+1} - \hat{\beta}_t' x_{t+1}.$$

The standardised recursive residuals $\tilde{v}_{t+1} = v_{t+1} / \sqrt{f_{t+1}}$ are $\stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ under the null of a correctly specified model.

9.3.2. CUSUM and CUSUMSQ tests

CUSUM: $W_t = \sum_{j=k+1}^t \tilde{v}_j / s$. Under parameter constancy $E(W_t) = 0$; divergence from zero signals a break in the mean.

CUSUMSQ: $S_t = \sum_{j=k+1}^t \tilde{v}_j^2 / \sum_{j=k+1}^T \tilde{v}_j^2$. Under constancy $E(S_t) = (t - k) / (T - k)$, a straight line from 0 to 1. Departures signal variance instability.

10. Information Criteria for Model Selection

Let $k = p + q + 1$ be the number of estimated parameters and T the sample size. The three standard criteria are:

$$\begin{aligned} \text{AIC} &= \ln \hat{\sigma}^2 + \frac{2k}{T}, \\ \text{BIC} &= \ln \hat{\sigma}^2 + \frac{k \ln T}{T}, \\ \text{HQIC} &= \ln \hat{\sigma}^2 + \frac{2k \ln(\ln T)}{T}. \end{aligned}$$

- **AIC**: not consistent; tends to select larger models.
- **BIC**: strongly consistent; heavier penalty for extra parameters; preferred in practice.
- **HQIC**: intermediate penalty; also consistent.

A 7×7 grid of ARMA(p, q) models with $p, q \leq 7$ is estimated for the first-differenced log USD/BRL exchange rate. BIC selects an AR(2) as the most parsimonious model with the lowest criterion value (-6.2108 at $(p, q) = (0, 2)$ after accounting for common factors). The two complex roots imply a pseudo-periodic pattern. CUSUM does not reject parameter constancy in the mean, but CUSUMSQ and the normality test reject — evidence of time-varying variance, motivating GARCH (Lecture 5).

11. Forecasting from ARMA Models

11.1. Optimal l -step-ahead forecasts

Define $X_{T+l|T} = E[X_{T+l} | \mathbf{I}_T]$. Forecasts are computed by the rule:

$$X_{T+l|T} = \phi_1 X_{T+l-1|T} + \cdots + \phi_p X_{T+l-p|T} + u_{T+l|T} + \theta_1 u_{T+l-1|T} + \cdots + \theta_q u_{T+l-q|T},$$

where $X_{T+l|T} = X_{T+l}$ for $l \leq 0$ and $u_{T+l|T} = 0$ for $l > 0$, $u_{T+l|T} = u_{T+l}$ for $l \leq 0$.

11.2. AR(1) forecasts

$$X_{T+l|T} = \phi^l X_T, \quad \text{MSE}(X_{T+l|T}) = \sigma^2(1 + \phi^2 + \cdots + \phi^{2(l-1)}).$$

The forecast reverts to the mean as $l \rightarrow \infty$. Under normality:

$$X_{T+l} | \mathbf{I}_T \sim N\left(X_{T+l|T}, \hat{\sigma}^2(1 + \hat{\phi}^2 + \cdots + \hat{\phi}^{2(l-1)})\right).$$

11.3. MA(1) forecasts

$$X_{T+l|T} = \begin{cases} -\theta_1 u_T & l = 1, \\ 0 & l \geq 2. \end{cases}$$

The MA(1) collapses to the unconditional mean after one step.

11.4. ARIMA(0,1,0) with drift (random walk)

$$X_{T+l|T} = X_T + l\mu, \quad \text{Var}(v_{T+l}) = l\sigma^2.$$

The forecast variance grows linearly with the horizon, producing characteristically wide prediction intervals.

11.5. Forecasts of transformed variables

When the model is fitted to $Y_t = \ln X_t$, the optimal forecast of the original series under normality is:

$$\hat{X}_{T+l|T} = \exp(Y_{T+l|T} + \frac{1}{2} \text{Var}(v_{T+l})).$$

11.6. Forecast evaluation metrics

Let $\{X_t\}_{t=T+1}^{T+h}$ be the realised values and $\{X_{t|T}\}$ the forecasts.

Metric	Formula
RMSFE	$\sqrt{\sum_{t=T+1}^{T+h} (X_{t T} - X_t)^2 / h}$
MAE	$\sum_{t=T+1}^{T+h} X_{t T} - X_t / h$
MAPE	$\sum_{t=T+1}^{T+h} X_{t T} - X_t / X_t / h$
Theil's U	$\frac{\sqrt{\sum (X_{t T} - X_t)^2 / h}}{\sqrt{\sum X_{t T}^2 / h} + \sqrt{\sum X_t^2 / h}} \in [0, 1]$

The MSE decomposes into bias proportion, variance proportion, and covariance proportion — useful for diagnosing systematic forecast errors.

12. Applied Example: USD/BRL Exchange Rate

The dataset `bolsa.xls` contains daily data from January 2000 to March 2003 for the Bovespa index, the IBX index, the BRL/USD exchange rate, and the S&P 500.

1. **Identification:** All series in levels exhibit slowly decaying ACFs consistent with stochastic trends. The ADF test fails to reject a unit root in the log exchange rate.
2. **Transformation:** $\Delta \log(\text{Dolar})$ is stationary. Its ACF/PACF suggests $p, q \leq 7$.
3. **Model selection:** BIC over a 7×7 grid selects AR(2) (two complex roots $\Rightarrow$ pseudo-periodic behaviour).

4. **Validation:** Recursive residuals show variance instability near observations 300 and 650. CUSUMSQ and normality tests reject.
5. **Forecasting:** The AR(2) is re-estimated on the sample excluding the last 7 observations. All 7 out-of-sample realisations fall within the 95% prediction intervals, but the bands are wide.

Violation of constant variance in the AR(2) residuals is the canonical motivation for ARCH/GARCH models. The Engle (1982) ARCH test on the squared residuals, and the GARCH(1,1) extension, are covered in Lecture 5.

13. Core References

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These lecture notes draw on `lecture1.tex` (stochastic process foundations and ARMA theory) and `IdentEstValid.tex` (identification, estimation, and validation of ARIMA models) by Prof. Pedro Valls Pereira, EAESP-FGV.